

Review of Current Trends in Sustainable Construction

Monika Zajemska ¹, Dorota Wojtyto ², Joanna Michalik ² and Szymon Berski ^{3,*}

¹ Faculty of Production Engineering and Materials Technology, Częstochowa University of Technology, 19 Armii Krajowej Ave., 42-200 Częstochowa, Poland; monika.zajemska@pcz.pl

² Faculty of Management, Częstochowa University of Technology, 19B Armii Krajowej Ave., 42-201 Częstochowa, Poland; dorota.wojtyto@pcz.pl (D.W.); joanna.michalik@pcz.pl (J.M.)

³ Faculty of Computer Science and Artificial Intelligence, Częstochowa University of Technology, 73 Dąbrowskiego Str., 42-201 Częstochowa, Poland

* Correspondence: szymon.berski@pcz.pl

Abstract: One of the most important trends in today's construction is energy efficiency. Nowadays, construction is undergoing a revolution thanks to modern technologies. Technological innovations not only speed up the construction process but also improve the quality, durability, and safety of the structure, ensure energy efficiency, as well as aesthetics and comfort of use. Modern technologies, such as photovoltaic panels and heat pumps, can produce their own electricity and heat for buildings' own needs with minimal use of fossil fuels. This not only leads to economic benefits, but also has a positive impact on the environment. Furthermore, choosing the right materials is the key factor in sustainable construction. Concrete, steel and other traditional building materials, although durable, have a negative impact on the environment due to the high energy consumption during their production. This study employed a SWOT-TOWS analysis to determine the strengths, weaknesses, opportunities, and threats of the basic construction materials used for sustainable building. Its purpose was to compile all construction materials and evaluate them using the same technological, environmental, and socio-economic indicators. The results of the analysis allowed for the comparison of materials in terms of their usability in sustainable construction and provided the opportunity to determine strategies for their further application, thereby filling the research gap in this area. Taking into account care of the natural environment, the article presents current trends in modern sustainable construction in the world. Using SWOT analysis, the advantages and disadvantages of sustainable construction were indicated, and the economic, social, legal, environmental, and technological barriers to their development were discussed. Particular attention was paid to the use of pro-ecological technologies that reduce energy consumption and greenhouse gas emissions.

Keywords: sustainable construction; building materials; energy saving; SWOT analysis



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1. Introduction

Sustainable construction can be defined as the efficient use of resources [1] with green design [2]. Modern housing developments are constructed sustainably, that is, in a way that benefits residents as well as the environment [3]. Sustainable building is not just a philosophy but a practical approach to creating residential and commercial spaces in harmony with nature [4]. At the heart of this approach is the belief that it is possible to build in a way that simultaneously meets needs and protects the environment, promoting the health and well-being of people [5]. It is important that the design, construction and use of buildings take into account aspects such as the use of efficient energy, environmentally friendly technologies, the reduction of water consumption [6] and the choice of building

materials that are both durable and environmentally friendly [7]. The idea of sustainable construction is to bring long-term benefits to the environment and to the current and future generations. The pursuit of the aforementioned goals is taken into account at the stages of designing, building and using the developments. Specific solutions are a very important part of sustainable construction, as they bring real benefits to residents and the environment. They are divided into two groups [8]: ecological [9] and well-being [10]. The group of green solutions includes renewable energy (e.g., photovoltaic panels) [11], light-colored roofing or roofs covered with extensive vegetation [12], rainwater collection systems, electric vehicle charging stations, solar benches, LED lighting, patios with greenery [8], etc. Equally important are other parameters ensuring a quality living environment for people [13], such as thermal and acoustic comfort [14], access to daylight [15], good ventilation, low levels of air pollution [16], and outdoor relaxation areas [17]. One of the pillars of sustainable construction is the building life cycle [18]. Buildings are to be designed so that they can be refurbished, recycled and remanufactured or returned to the production cycle [19].

To overcome today's social and environmental challenges, sustainable building requirements are needed right from the design stage; however, entrepreneurs, policy makers and technical professionals continue to ignore them [20]. Organisational strategies, the level of sustainability maturity and the ability to change corporate values and beliefs play significant roles in this key movement. At the same time, external stakeholders, such as investors, banks, communities, suppliers, regulatory agencies and insurance companies, can impose certain behaviours on the market and influence change. In addition, organisations should build their sustainability strategy and implement sustainable project management [21]. The article is of a review nature. In its construction, in addition to the study of the subject literature, a number of other research methods, techniques, and tools were also used. Their application was two staged, initially involving the processing of materials and subsequently the systematisation of the collected results. To achieve the above objectives, the following research methods were used: comprehensive SWOT-TOWS analysis, analysis and synthesis, abstraction, organisation, grouping, comparisons, induction and deduction, interpretation, definition, and selection. The work also utilised the VOSviewer program, version 1.6.20, a spreadsheet for SWOT calculations, and graphic software. A detailed analysis was conducted on the literature related to the research subject, including monographs, scientific publications, and online resources. The issues discussed in the article are still open topics and require further research in this area, including the use of other (empirical) research methods. In general terms, the SWOT-TOWS analysis was conducted based on the literature data. The main objectives of this study are outlined below.

- presentation of current trends in sustainable construction and the direction of their development;
- indication of the main barriers to the development of sustainable construction in relation to potential benefits;
- comprehensive comparison of selected building materials (concrete, steel, wood, glass) in terms of their use in sustainable construction;
- indication of the strengths and weaknesses of individual construction materials as well as the opportunities and threats of their application in sustainable construction;
- based on the conducted SWOT-TOWS analysis, identification of the most advantageous construction materials in the context of applications in sustainable construction;
- indication of development strategies for individual construction materials in current socio-economic conditions.

Based on the current state of knowledge and the specified objective and research problem, the following research hypotheses have been formulated:

- It is assumed that the trend in sustainable construction is moving towards an increasing use of various technologies and the integration of innovative solutions with traditional ones, in order to enhance, among other things, energy efficiency and the use of recycled materials. The future of construction may be shaped by designs that incorporate the use of technologies, such as solar panels and smart energy management, as well as the introduction of green architecture (e.g., green roofs, rainwater management systems). These actions indicate the growing need to strive for more sustainable development in construction practices.
- It is assumed that the main barriers to sustainable construction are technological barriers. However, the main benefits of applying sustainable construction are ecological, and, consequently, also social.
- It is assumed that materials, such as metals (aluminium and steel), plastics (polycarbonate), and composites (e.g., carbon fibre, ceramic materials) have the greatest technological utility. The greatest ecological benefits are biodegradable materials (bioplastics), secondary materials (glass, paper, metal), and wood. However, the greatest social utility is recycling materials and those with low energy consumption in production (some types of ceramics).
- It is assumed that in the context of sustainable construction, wood and glass have the most strengths, while wood has the most threats.
- It is assumed that the material with the best properties in terms of the studied factors for sustainable construction is glass.
- It is assumed that an aggressive strategy concerns glass as a construction material, with strong expansion of opportunities and strengths.

2. Theoretical Background

Statistical Overview of Sustainable Construction

This paper includes a comprehensive literature review of sustainable construction using the popular science mapping tool VOSviewer [22]. The analysis was conducted using the Scopus database for the following keywords: “sustainable construction, wood, cement, steel, glass” (Figures 1–3). Similar analyses for sustainable construction have been conducted by a number of researchers, including Araújo et al. [23], Zhang [21], S’anchez-Garrido [24], Shashi et al. [25], and Kazemi et al. [26]. The analysis shows that the issue of sustainable construction is relatively new, and in the analysed years of 1996–2025, the vast majority of publications [27] are from the last ten years, with cement dominating as a building material [28].

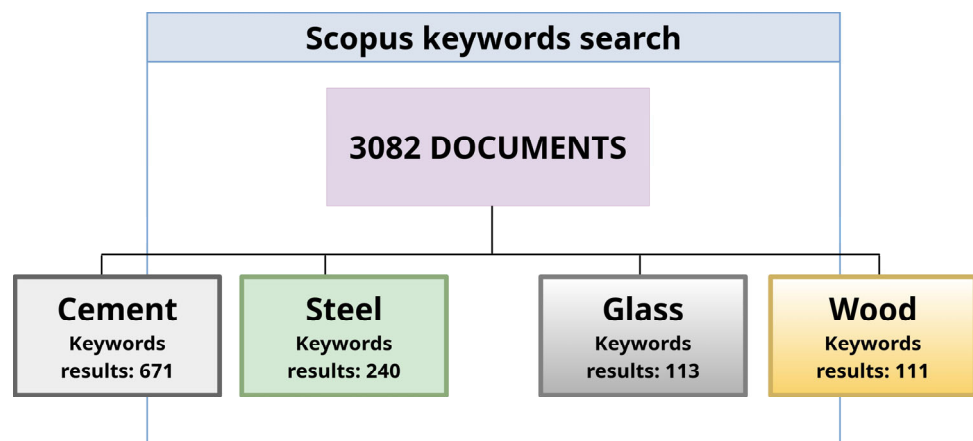


Figure 1. Schematic of keyword search and refinement in Scopus.

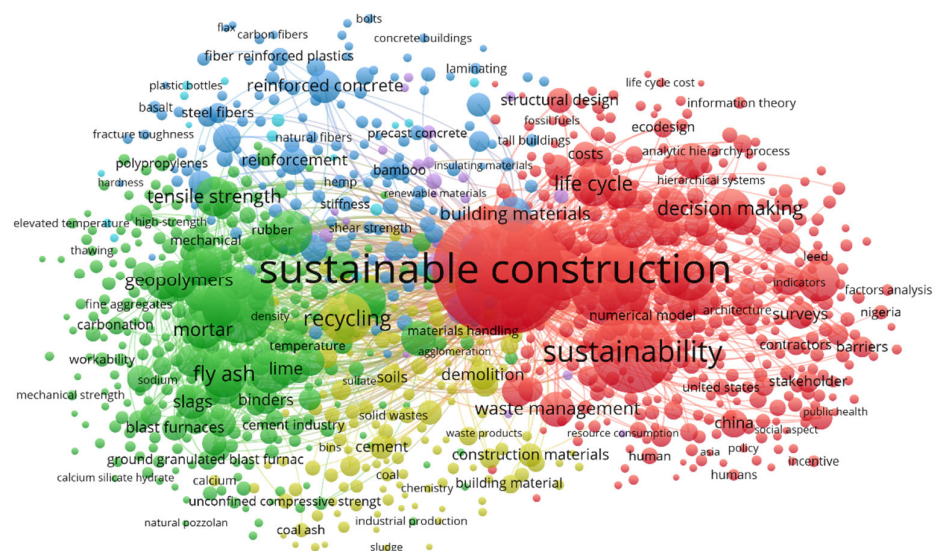


Figure 2. VOSviewer network visualisation of terms associated with sustainable construction.

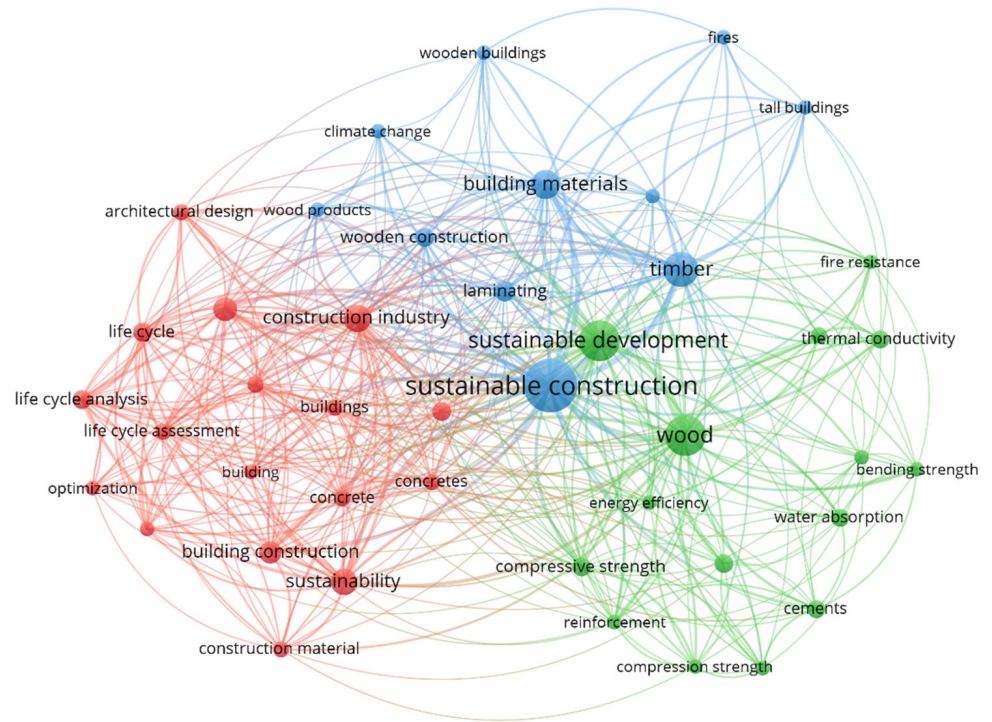
Bibliometric analysis has shown that the issues addressed in the article are current and scientifically significant, as evidenced by the high level of interest from researchers and numerous publications. At the same time, there is a research gap in areas such as the use of wood in sustainable construction, which has not yet been sufficiently explored, making the chosen topic promising for research and worthy of further investigation and broader analysis.

The co-occurrence of the studied keywords is shown in the form of connection maps. This was performed by finding keywords that showed concepts that appeared at least five times. The connection maps between the keywords were used to identify the main research clusters, marked in different colours in Figures 2 and 3.

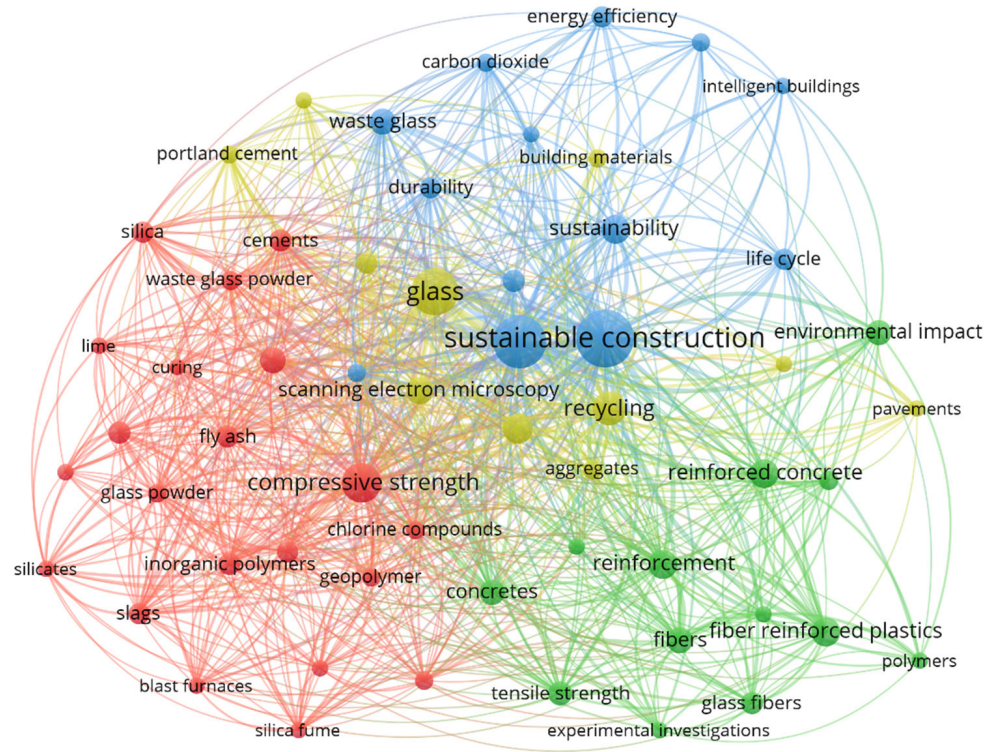
The conducted analysis of keyword co-occurrence showed that the majority of research clusters concern the issue of cement use in construction, while the fewest clusters were identified for wood, specifically, only three. The individual clusters, marked in different colours, represent groupings connected by a common feature, namely the keyword “sustainable construction”. The size of the point in Figures 2 and 3 indicates the frequency of keyword occurrences; the larger the point, the more frequently the term appears. As the analysis shows, in each case examined, the dominant term is “sustainable construction”.

When analysing Figure 2, one can observe a dense network of connections between the keywords. Marked in red, the largest cluster mainly concerns the area of sustainable construction in the context of waste management in the construction industry and the life cycle of materials. The second dominant cluster is the one marked in green, primarily encompassing aspects related to recycling and the use of production waste, such as “fly ash”, in the production of building materials. The network of connections for individual building materials presented in Figure 3 showed that cement and steel are characterised by the densest and most complex network of connections, with clearly distinguished clusters, indicating the frequency of occurrence of these keywords in scientific publications. A significantly less dense network of connections is observed for glass and wood, as evidenced by the smaller number of clusters and the size of the points.

As already mentioned, the most common material used in construction is cement. However, the use of construction materials has seen the introduction of innovative solutions over the past few years. This article undertook actions aimed at systematising various groups of materials.

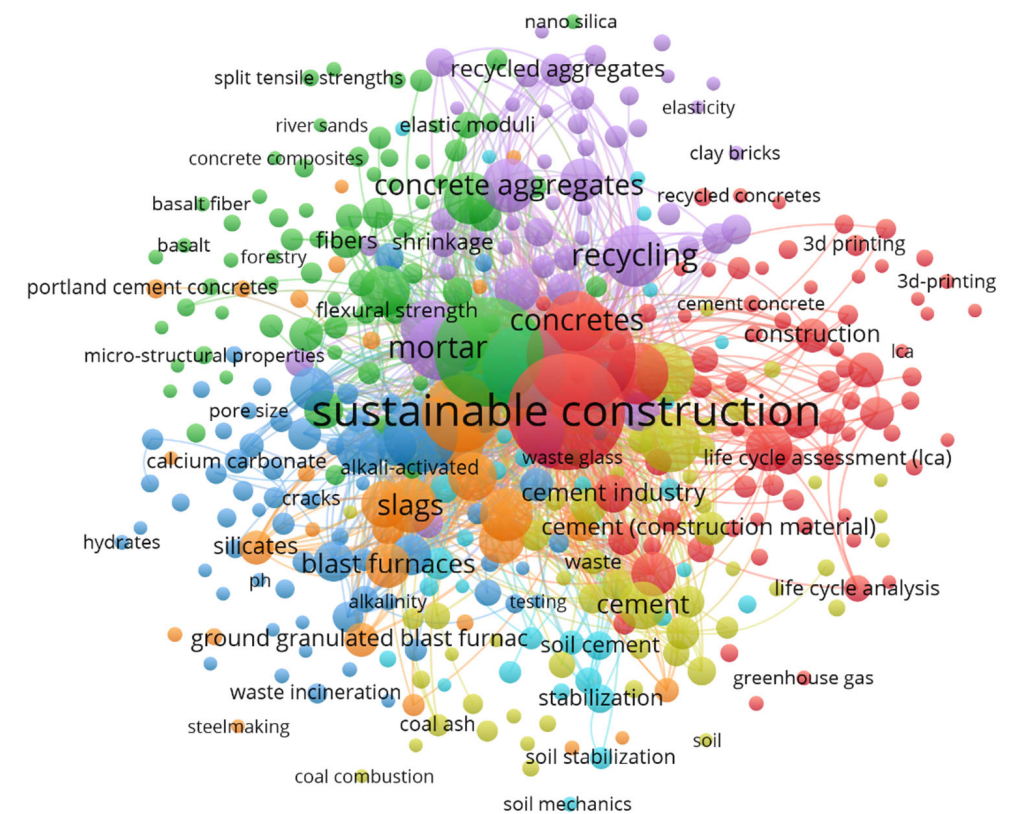


wood

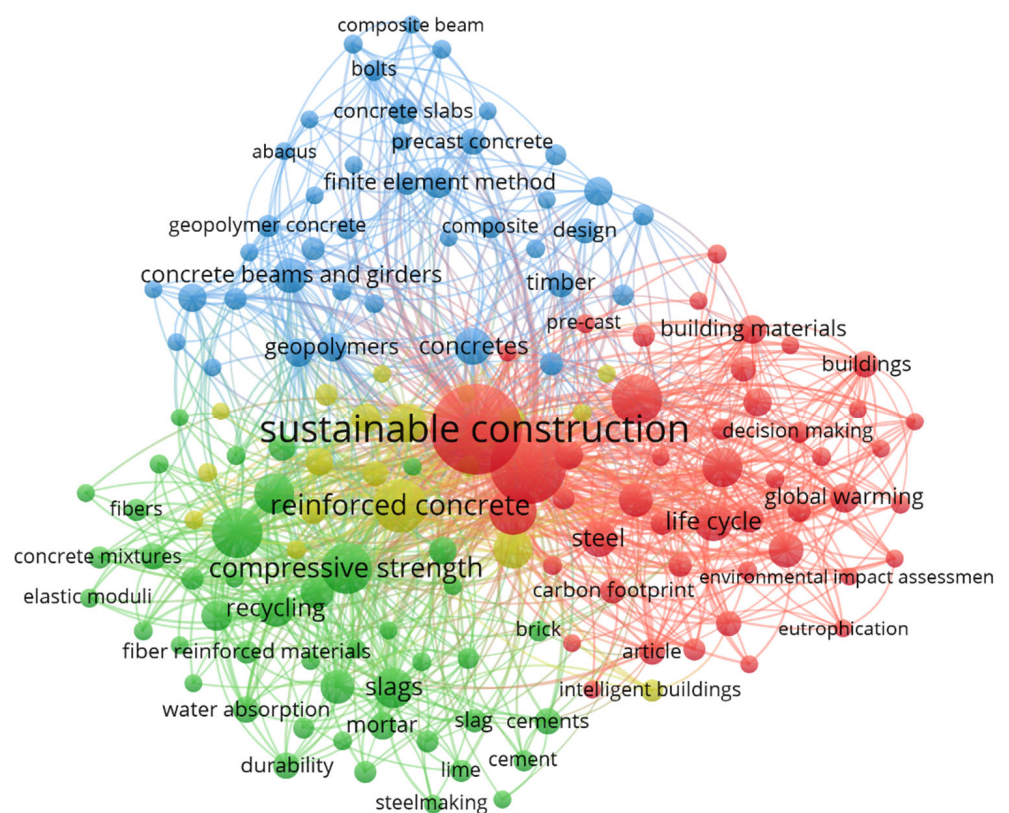


glass

Figure 3. Cont.



cement



steel

Figure 3. VOSviewer network visualisation of terms associated with sustainable materials.

3. Innovative Solutions in the Use of Building Materials

The materials used in sustainable construction can be divided in several ways. One common division is to distinguish three groups of materials:

1. Renewable materials can be included in the first group. These include materials of natural origin that can be renewed in a relatively short period of time. These include wood, bamboo, straw, mineral wool, flax, hemp and others.
2. Recycled materials: These materials are recycled in waste processing, making it possible to use resources that already exist and reduce waste. Examples include steel [29], recycled concrete [29], recycled glass [30], recycled plastics and others.
3. Materials with low environmental impact: This group includes materials that have a lower overall environmental impact than traditional building materials. These may include, for example, low-carbon materials (e.g., concrete with cement replacements [31]), low-embodied energy materials, such as stone, straw or burned wood, or materials produced in a more environmentally friendly manner [32]. This group also includes traditional materials that have been subjected to innovative solutions.

Innovative solutions in the use of the traditional building material of cement include the following:

- The use of cementitious materials containing iron-rich industrial waste (IWP) to support a nanophotocatalyst, such as titanium dioxide (TiO_2), to produce a visibly active Fe— TiO_2 composite exhibiting antibacterial properties [33].
- The use of recycled powder (RP) obtained from construction waste [34].
- The use of lithium slag (LS) as a precursor to geopolymer concrete, geopolymer technology uses alkali activators and aluminosilicate-rich materials, provides less environmental impact and shows performance comparable to traditional cement-based concrete, and most importantly contributes to a closed-loop economy in the construction industry [35].
- The use of recycled coarse aggregates (RCA) and construction and demolition waste (C&D) in sustainable concrete production [36].
- Cementitious mixtures, so-called “three-component” mixtures, consisting of mineral admixtures, limestone and clinker [37].
- The use of bauxite residue, commonly known as red mud, a byproduct of the Bayer process used in aluminium extraction, in concrete-based applications [38].
- The use of eggshell waste as a material to partially replace cement in clay and cement-based building materials; eggshell waste is among the most common agricultural waste from food processing plants [39].
- The use of various slags, including furnace slag, electric arc furnace slag (EAFS), steelmaking slag (SFS) and ground granulated blast furnace slag (GGBFS) in the development of sustainable building materials [2].

The important cement replacements are geopolymers, which are primarily used in the production of concrete containing a binder formed on the basis of aluminosilicates instead of classic cement. The main difference between the two binders is the type of chemical reaction that causes them to harden and turns the plastic concrete mix into a solid. Furthermore, geopolymer concretes demonstrate high compressive strength, very low shrinkage and low creep, as well as high resistance to acid and sulfate corrosion. The apparent advantage of geopolymer binder over OPC Portland cement is certainly the low carbon footprint, rapid strength gain and its high values. In practice, the use of geopolymer concretes is still very limited due to their high price, which is determined by the use of relatively large amounts of sodium hydroxide and aqueous silicate solutions [9,40,41]. There are many varieties of

geopolymers, and recent studies have revealed that the incorporation of GO up to 1% can result in enhanced mechanical strength, durability, and reduced environmental impact.

The use of recycled glass (cullet), instead of sand and aggregate, is a solution to the shortage of river sand, which has so far been used as an ingredient in the concrete mix. In addition, finely ground glass has pozzolanic properties, which means that by binding with hydroxide, it improves the strength and durability of concrete and, by partially replacing cement in the mix, indirectly reduces CO₂ emissions during cement production. An additive described in both Polish (PN-EN 15167-1:2006 [42]) and American (ASTM C989 [43]) standards is ground granulated blast furnace slag (GGBFS)—a byproduct of iron and steel production that can be ground into a fine powder and used as a cementitious material in concrete mixes [44]. The additive can improve the workability, strength and durability of concrete, and can reduce its heat of hydration, making it suitable for mass concrete applications. Materials such as geopolymers, fly ash, and glass are used in the latest technologies in 3D concrete printing (3DCP) that reduce costs, materials, and time, increase safety and minimise environmental pollution. In addition, 3D printing makes it possible to create complex and intricate designs [45] that were previously difficult to achieve with traditional methods, particularly in areas such as precast concrete project analysis [46], the study of the competitiveness of construction projects [47], cloud computing for sustainability of small construction projects [48] or modelling digital twins [49]. The papers [50,51] present selected CO₂ capture technologies, such as amine scrubbing, calcium looping, oxyfuel combustion and direct separation, and their use in the production of concrete materials.

One of the many green alternatives aiming to replace structures made of energy-intensive materials [52] is modern wooden structures [53]. This type of construction is now very popular around the world [54] and is already part of the standard of living in many countries [55]. The potential inherent in the development of wood-based construction is currently not being fully exploited, so it is necessary to focus more on research and innovation in the wood construction sector (Figure 4).

Current research conducted around the world addresses issues in the scope of innovation, indicating possible aspects of innovation for the construction sector and the innovation potential of various wood construction systems. In the work [53], the parameters of wood as a construction material were presented, the most important of which is “construction time” (5.6), which is positively correlated with “price” (4.9). The next important parameter is “thermal and technical properties” (5.4), while the least important parameters are “environmental footprint” (4.5), “energy balance during use” (4.5), and “lifetime” (4.4). The less important are “acoustic properties” (3.6) and “fire resistance” (3.2). In addition, the authors of the article [53] examined, from the point of view of the respondents, the advantages and disadvantages of wood as a construction material. The best rated were “impact on the environment” (100% of the respondents considered it an advantage), “renewable material”, “construction time”, “thermal-technical properties” and “energy balance” (according to the order of occurrence from 97% to 94%). The “price” parameter was considered an advantage by 72% of the respondents. The worst rated were “lifetime”, “acoustic properties” and “fire resistance” (from 67% to 52%). To sum up, wooden structures have great potential and are appreciated for their ecological character, lightness and speed of assembly; however, there are still concerns related to durability, acoustics and fire resistance. The market appears to be open to the development of this technology, although it requires a change in perception, legal support and technological innovations.

In addition to the standard uses of wood in construction, such as load-bearing structures, walls and ceilings, roofs, facades or interior finishes with wood, technologies are emerging to replace typical load-bearing structures in RES technologies, such as for wind

turbines [56]. The effect of using such materials is a reduction in CO₂ emissions i.e., significantly lowering the carbon footprint [37] associated with the investment [57].

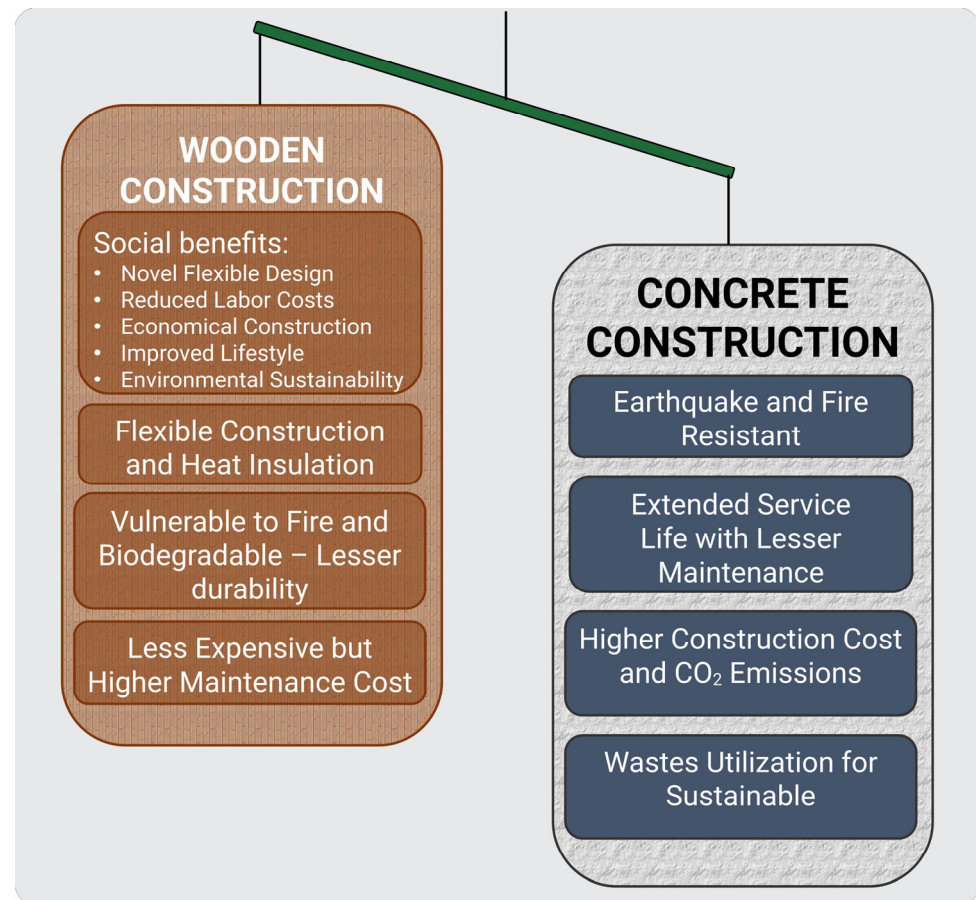


Figure 4. Advantages of wood construction over traditional construction [55].

In addition, when selecting materials in sustainable construction projects, it is important to take into account LCA and BIM, i.e., different levels of use of technologies for digital recording of physical and functional properties of a building, in parametric form. As part of the improvement of tools to support design processes, coupled with analysis methods such as LCA-BIM, are proposed [29].

The sustainable building initiatives that are underway around the world include the following:

- use of insulating biomaterials, such as products available in the region, i.e., agricultural waste [58], such as grain straw, hemp and olives [7];
- improving the efficiency of energy-efficient management of industrial buildings through the use of artificial intelligence [59];
- optimising [60] and retrofitting [61] existing buildings for energy conservation and decarbonisation throughout the life cycle [62];
- cool roofs and airflow-enhancing technologies can also be mentioned—they are efficient, passive cooling methods that reduce energy demand and improve comfort for building occupants [63].

Systematising and reviewing building materials with environmental aspects in mind are important topics that are gaining importance in the context of sustainable development and green building. It requires a holistic approach that takes into account the entire life cycle of materials and their impact on the environment. Promoting sustainable practices in

construction can help reduce the negative impact on our planet and improve the quality of life of residents. The next section of the article presents several key points on this issue.

4. Environmental Aspects of Sustainable Construction

The path to sustainability is largely limited by the practices of the construction industry [11], which is a significant generator of waste and other anthropogenic emissions [64]. It follows that shifting to cleaner technologies and construction methods [65] by implementing innovations that minimise consumption of resources and anthropogenic pollutants, such as greenhouse gases (GHG), aerosols, carbon dioxide, sulfur oxide, nitrogen oxide, benzene and other emissions that adversely affect humans and the environment, will promote the achievement of the Sustainable Development Goals [66]. To ensure sustainability, it is imperative that the construction industry [67], as a centre of economic development, seek solutions that are cost-effective, pragmatic and inclusive [68]. There are many aspects through which modern construction supports sustainability and has a positive impact on the environment. The use of appropriate green technologies reduces greenhouse gas emissions [69]. This is influenced, among others, by the reduction in energy consumption in sustainable buildings, good insulation performance and other elements that characterise green construction [54]. In an era of shifting away from dependence on fossil fuels, the use of renewable energy sources is of utmost importance. As mentioned earlier, what is key to sustainable construction and sustainable development is the reduction in energy consumption through, among others, the efficient use of energy [18]. The essence is to design the building so as to provide maximum comfortable thermal conditions [70] with minimum energy consumption. In order to achieve such an effect, it is necessary to ensure adequate thermal insulation, airtightness of the building, eliminate thermal bridges and bring as much green energy into the building as possible. The energy efficiency of a sustainable building also depends on the entire life cycle of the building, not just the time of its target operation [71], beginning with the construction of the building, through its upkeep and maintenance, to its demolition [72].

Along with socioeconomic ones, the environmental aspects [73] are one of the main groups of indicators for sustainability of the construction sector [74].

Noteworthy is a study by Koengkan et al. [8], which examined the impact of environmentally friendly homes on the economic development of the Lisbon metropolitan area. In their study, they analysed data from 18 municipalities over the period of 2014–2020, using regression analysis with ordinary least squares (OLS) and fixed effects. The results indicate that national policies promoting residential energy efficiency have a positive impact on economic development. Policies, such as grants, loans and tax credits, encourage homeowners to invest in energy-efficient technologies, increasing household disposable income and contributing to economic growth. The study found that an increase in the number of new buildings has a positive impact on economic development, leading to job creation and increased demand for construction materials and services. In addition, building environmentally friendly homes can reduce energy consumption by lowering energy costs for homeowners and businesses, and ultimately stimulating economic growth.

In the next part of the article, a SWOT analysis for construction materials was conducted, aimed at helping to understand the strengths and weaknesses, as well as the opportunities and threats associated with the construction materials market.

5. SWOT-TOWS Analysis for Construction Materials

SWOT-TOWS analysis was chosen for its practicality, universality, and ease of assessment. It is a comprehensive method because it presents all internal and external factors influencing the developmental potential of a given area in a single matrix. SWOT-TOWS

allows for focusing attention on the most significant strategic development factors. A significant advantage of this method is also its modern perspective on the given problem. The disadvantage of this method is the subjective assessment in selecting factors and determining their weights and significance. It depends on the state of the knowledge possessed and/or access to information.

The SWOT method is a very popular tool, but it is not without its limitations. Insufficient data, frequent lack of objectivity from researchers who are unable to properly assess strengths and weaknesses, or, for example, underestimating threats while focusing on strengths and opportunities—these are some of the issues that should be taken into account when using this method.

When preparing a SWOT analysis, it is also necessary to consider legal constraints that can sometimes affect the interpretation of research results, such as data protection and legal liability, as well as laws arising from violations of fair competition or intellectual property rights (which are not the subject of analysis in this article).

Due to growing concerns about environmental issues and resource and energy consumption in the construction sector, the development of “green buildings” which, as they are defined, result in a reduction in the negative environmental and human impacts, is becoming increasingly important. However, the escalation of sustainable construction faces a number of entry barriers and obstacles at the implementation stage. Undoubtedly, the first basic method useful in the broad analysis of sustainable construction in terms of its positive and negative aspects is benefit–cost analysis [75]. Various methods—both objective and subjective ones—are used for estimating these two quantities. Unfortunately, they are often subject to measurement error or a lack of proper conditions for conducting the study, which also results in a wide variation in the evaluation of the costs and benefits of sustainable construction in general. According to a study conducted by [76], the ambiguity of actual costs and benefits is a major obstacle to the development of green architecture. It should also be taken into account that, for example, green buildings, the costs that matter are different for investors than for unit owners or tenants. Construction costs are important for investors, while ongoing costs are important for users. Among others, this is the basis of the difficulty with specifying these figures. Methods of economic and non-economic analyses used in science and practice that seek to answer this nagging question of profitability and return, or the correlation between green strategies and green performance, also include, for example, interviews, surveys, statistical and economic analyses, experiments, and simulations, benchmarking, risk analysis, technological and ecological research, and other engineering studies [75].

Based on cost–benefit analysis and on research using the above methods, as well as theoretical considerations derived from this work, the advantages and disadvantages of sustainable construction, both measurable and non-measurable, can be presented in general terms. Measurable aspects provide an opportunity to quantify the impact of implementation of sustainable construction intentions. They concern indirect (long-term) benefits and costs as well as direct (short-term—immediately perceptible) ones. Due to the difficulty with accurately estimating the benefits and costs of sustainable construction, non-measurable aspects are also included in this discussion. They are not subject to quantitative estimation and are primarily concerned with the general and social scope. Accordingly, Table 1 shows the measurable advantages and disadvantages, while Table 2 shows the non-measurable advantages and disadvantages of sustainable construction.

Table 1. The measurable advantages and disadvantages of sustainable construction.

Measurable	
Advantages	Disadvantages
▪ Reducing and reusing waste. ▪ Reducing the number of pollutants and the degree of environmental degradation. ▪ Reducing greenhouse gas emissions. ▪ Conservation of natural resources. ▪ Caring for the environment—use of safe materials that do not emit harmful substances into the atmosphere; long-term environmental benefits. ▪ Implementing water conservation strategies, protecting the ecosystem. ▪ Energy savings in production of materials, as well as installation and operation. ▪ Use of ecological solutions—energy-efficient LED lighting, use of photovoltaic panels, thermal efficiency improvement of walls and windows, rainwater retention, stand-by elevators with energy recovery systems, use of daylight, heat pump, recuperation (air purification) and HVAC (Heating, Ventilation, Air Conditioning) systems. ▪ Energy use, including increasing energy use efficiency through installation of windows with high thermal efficiency, adherence to standards for obtaining insulation parameters of walls, ceilings, roofs, and seals, installation of solar panels, especially for solar water heating and even electricity generation, installation of external blinds, construction of vestibules, use of groundwater heat for heating buildings, and recuperation systems for ventilation of buildings with heat reuse. ▪ Thermal and acoustic comfort, natural ventilation. ▪ Reduction in natural hazards. ▪ Smart energy management (building automation systems (BMS)) by monitoring and regulating energy consumption in real time. ▪ Heat recovery (recuperation, solar panels, photovoltaics, low-carbon materials, as well as remote control systems for all installations in the building). ▪ Thermal comfort (passive buildings) without using a heating system or air conditioning. ▪ Efficient use of energy and other natural resources. ▪ Limiting water consumption and ensuring adequate water quality. ▪ Reduced water loss through reuse (rainwater, gray water) and use of water from washing machines and dishwashers, after treatment, can be used to irrigate green spaces, or, if treated, for other purposes, such as washing cars. ▪ Lower CO₂ emissions.	▪ Small number of companies in the market that undertake green building contracting activity. ▪ Lack of universal design and construction methods. ▪ The construction costs of green buildings are higher than those of the traditional method of building. However, these investments pay for themselves within 5–15 years, on average 6 years. ▪ The cost of passive houses is 10–20% more than traditional houses (in Poland). ▪ High upfront costs during the design and certification processes. ▪ Ambiguity of actual costs and benefits—significant differences in methodologies of cost assessment, soft costs (upfront fees related to design) and hard costs of construction, building materials and services, lack of clear and systematic studies of design and construction costs. ▪ Higher construction costs than in the case of conventional buildings (41% of green buildings are more than 11% more expensive than similar conventional buildings). ▪ Erroneous economic thinking—the lowest possible construction costs result in the best energy efficiency (it is the other way around). ▪ Custom designs are much more expensive than standard ones, and there are now several companies on the market that offer ready-to-buy passive houses on a specific plot of land. ▪ New products on the market continue to be much more expensive (e.g., windows and doors consistent with the passive standard). ▪ Fluctuations in prices of materials. ▪ Green roofs pose fire hazards, higher maintenance requirements, shading of the photovoltaic panels due to excessive plant growth, the effects of the photovoltaic modules on green growth, the effect of improper panel height on air circulation inside the building. ▪ High costs of upgrading existing buildings. ▪ Inadequate or no subsidy programs for low-energy or passive construction, e.g., in Poland. ▪ The duration of green building projects and their construction is significantly longer compared to non-green projects. ▪ List prices of buildings may differ from final prices due to the still quite poor knowledge of the market. ▪ Green architecture suffers from a lack of quantitative cost–benefit studies to accelerate its application and use.

Table 1. *Cont.*

Measurable	
Advantages	Disadvantages
- Lower taxes on CO₂ emissions. - It is possible to use both renewable materials (wood, bamboo, straw, wool, flax or hemp), secondary materials (concrete, glass, plastics) and materials with low environmental impact (e.g., concrete with cement replacements). - Bonus for LEED (Leadership in Energy and Environmental Design) certification. - Good Indoor Air Quality (IAQ). - It covers the entire life cycle of a building (closed loop). - Lightweight structures (such as those made of Neopor) can be erected in areas where traditional buildings could not be built. - Using the same material simultaneously for insulation and structural purposes (e.g., Neopor)—a technology both cheap and fast (takes less time to build). - Reducing heating costs; green buildings save between 30 and 50% of energy. - Reducing water consumption by about 30%. - Reducing emissions of harmful gases into the atmosphere by nearly 40%. - Higher employee productivity. - Less employee sick leave. - Long-term financial benefits for companies. - Lower operating costs (related to water, energy consumption)—lower electricity and water bills. - Higher property values. - Designing buildings in a flexible way that adapts to climate change. - High degree of implementation of energy-efficient and passive construction in Western Europe (especially Germany and Austria) due to research and development work. - In Western Europe, the cost of building passive houses is only a few percent higher, combined with a quick return on investment. - Development of the energy-efficient construction sector—energy-efficient solutions have become available through a variety of support instruments, including for the average investor. - Opportunities to subsidise the construction of green buildings (subsidies, loans, support programs—e.g., solar panels, PV, building insulation, etc.). - Benefits of 3D concrete printing technology in construction (construction efficiency, reduction in construction waste, increased energy efficiency, flexibility and speed of design, reduction in work accidents and thus improvement of work processes themselves).	- Unreliable data collection sources, which may result in biased survey results, lack of data enforcement tools. - High cost of research and testing processes. - In many countries, such as Poland, there is lack of technology and scarcity of indigenous materials, forcing the import of more expensive materials from Western Europe. - Impediments related to specific location requirements for a home with very low energy consumption. - Complicated procedures involved in building an on-site wind turbine. - Lack of precise operating criteria and procedures; wrong use of standard construction technologies for green construction. - Lack of knowledge of passive house construction rules and contractor qualifications, due to poor knowledge of the market. - Lack of effective government incentives, subsidies and promotions. - Uncertainty as to the demand for and supply of products. - Monopoly on the technology used by a few companies - The claim that modular homes have low market value. - Incompetence of designers, manufacturers and component suppliers. - Limited number of contractors specializing in precast concrete systems. - Lack of early consulting on production and construction. - Design limitations due to transportation constraints. - Lack of a one-size-fits-all design and construction tool for Modular Integrated Construction (MiC). - Problems of financing energy-efficient construction. - Strong emphasis on lowest bid price rather than best value. - Difficulty in achieving economies of scale and returns on high initial investment. - Difficulties in obtaining funding for MiC projects. - High logistics prices, high bid prices for contractors, prohibitive costs of repair and rework for MiC. - Less demand for new construction; aging population and impoverishment. - Overinvestment or underestimation of the cost of sustainable construction due to incomplete knowledge of investors. - Limited access to components or their high price.

Table 1. *Cont.*

Measurable	
Advantages	Disadvantages
- Along with green construction, a smart home is also crucial, as it allows control of heating, lighting, ventilation, blinds, alarm and other installations in the building. This allows for a corresponding reduction in energy consumption and, consequently, expenses. - Green roofs, walls, facades—mitigation of urban heat islands, reduction of air pollution with fine particulate matter, they use 2.2–16.7% less energy and reduce rainwater flow by ca. 4%, reduce indoor temperature by up to 3.6%, increase oxygen production, absorption of pollutants, water retention, living space for insects, good noise attenuation, the gap between photovoltaic panels and the green roof increases the efficiency of the system by using biomass. - In modern trends, investors are more likely to opt for designs for smaller buildings, which allows for easier maintenance, lower costs for interior finishing, heating and reduced electricity consumption. - Plus-energy houses, i.e., houses that not only show low operating costs, but also produce surplus energy. - Efficient building evaluation systems: scientific verification systems that evaluate buildings based on a range of environmental and sustainability-related criteria, e.g., the U.S. LEED (Leadership in Energy and Environmental Design), or the U.K. BREEAM (Building Research Establishment Assessment Method). - Return on investment in 5 to 15 years, with an average of 6 years. - Less dependence on imports of energy commodities. - Addressing the energy resource deficit for future generations. - Fewer experts in building maintenance and operation. - Designing for fire protection and protection against crime—measurable security benefits. - Increased interest due to adoption of the Energy Efficiency Act. - More opportunities for sustainable construction outside the city in single-family housing, due to population migration from the city to the countryside.	- Difficulties in renovating, converting, or constructing buildings in the city in terms of sustainable construction (e.g., historic buildings). - Concentrations of population in one place (e.g., city centres). - Rising real estate prices and other macroeconomic figures (unemployment, inflation). - Aging population and divergent economic status of the population, resulting in different housing preferences.

Source: own research based on [32,75–81].

The identified advantages and disadvantages of sustainable construction were listed in the form of benefits and barriers, divided into technological, ecological and socio-economic factors, as shown in Figures 5 and 6.

Table 2. The non-measurable advantages and disadvantages of sustainable construction.

Non-Measurable	
Advantages	Disadvantages
■ Preserving the green landscape. ■ Improved quality of work/living environment and work/life comfort. ■ Improving the well-being and health of users. ■ Clean air, among others thanks to recuperation. ■ Green roofs, walls and facades—the psychological impact on bringing people closer to harmony with nature—biophilia, green roofs resemble parks, give positive esthetic reactions, green color calms down, improves biodiversity in the environment. ■ It meets the tenets of “ideal well-being”. ■ Construction is based on a location away from dense development, and close to recreational areas ■ It creates new trends in construction in line with ecological and environmental protection. ■ Changing people’s lifestyles to be environmentally friendly. ■ Balance in the city’s ecology. ■ Social responsibility for environmental protection. ■ Advancing social equality. ■ Creating a healthy environment indoors as well. ■ Increasing environmental awareness among green building users. ■ Participation in the design of accessible and safe public spaces. ■ Active involvement of local communities in the design/construction process. ■ Promoting better health and well-being of the community. ■ Fostering greater community involvement through knowledge transfer and access to information. ■ Esthetics and functionality of construction. ■ In many countries, e.g., Poland, good natural environment conditions for erecting and maintaining passive buildings.	■ A relatively new industry, considered a niche until recently. ■ The mental barrier of investors and users who are not convinced about innovative solutions in the scope of construction. ■ Resistance of the community to change and innovation, as well as resistance from potential customers, conservatism and skepticism. ■ Lack of public confidence in sustainable construction. ■ Conservative industry bias in favor of conventional construction. ■ Dominance of conventional construction practices/processes. ■ Concerns about the adaptability of construction and its flexibility to meet the needs of multiple generations. ■ Concerns about urban heritage or urban identity. ■ Monotonous type of construction and architecture e.g., MiC—Concerns about architectural creativity. ■ Lack of sufficient cooperation between scientists and R&D specialists with practitioners, i.e., architects, designers, building material manufacturers and investors. ■ Spreading of myths about green buildings. ■ Lack of sufficient factual and legal know-how; lack of dissemination of good construction practices, lack of knowledge transfer, insufficient education of the public, lack of knowledge and information flow among representatives of science, the industry and the user. ■ Poor public acceptance due to suspicions about meeting quality expectations. ■ Inadequate policies and regulations in this scope. ■ Lack of educational programs on structural and architectural aspects. ■ Stereotypical thinking, no major experience in sustainable construction. ■ Lack of employee competence in sustainable construction, resulting in complaints and disputes. ■ Barriers caused by urban planning of cities and suburbs. ■ A choice between the advantages of the environment (the well-being of the city) and energy-efficient housing.

Source: own research based on [75–81].

As can be seen from Tables 1 and 2, this study identifies the most measurable advantages and non-measurable disadvantages of sustainable construction, in general. This means that in a holistic approach to the implementation and execution of “green construction”, the benefits far outweigh the costs. Nevertheless, a large number of socio-economic and technological barriers mean that this type of construction is still in its early stages. This

is primarily due to the fact that the sustainable construction sector is still a new industry that the public is not fully convinced about, due to high initial costs, as well as a lack of knowledge and an information gap. However, analysing Tables 3 and 4, it can be concluded that the dominant benefits relate primarily to ecological aspects and, consequently, social aspects. The scarcity of natural resources and the idea of a healthier and better life in a clean environment prevail over the stereotypical thinking of conventional construction as the only right and sustainable approach in this day and age. No environmental barriers to sustainable construction were noted.

Technological	Ecological	Socio-economic
• Use of technological green solutions. • Thermal and acoustic comfort, natural ventilation. • It covers the entire life cycle of a building (closed loop). • Lightweight structures (e.g. made of neopor). • Using the same material for both insulation and construction purposes. • Use of better quality materials, including insulation better than traditional construction. • Designing buildings in a flexible way that adapts to climate change. • Benefits of 3D printing technology for concrete in construction. • Smart home development. • Plus-energy houses. • Less dependence on imports of energy commodities. • Design for fire protection and crime prevention. • Fewer building maintenance and operation experts. • Development of the energy-efficient construction sector.	• Reducing and reusing waste. • Reducing pollution. • Reducing greenhouse gas emissions, especially CO₂. • Conservation of natural resources. • Caring for the environment. • Energy conservation and smart energy management. • Reduction in natural hazards. • Reduction in water consumption and ensuring adequate water quality. • Less water loss, and water recovery. • Ability to use materials that are renewable, secondary and with low environmental impact. • Good Indoor Air Quality (IAQ). • Green roofs, walls, facades. • Efficient building rating systems. • Increasing use and interest due to the entry into force of the Energy Efficiency Act. • Preserving the green landscape. • Clean air, among others thanks to recuperation. • Balance in the city's ecology. • Creating a healthy environment, including indoors. • Higher property value • High degree of implementation of energy-efficient and passive construction in Western Europe.	• Bonus for LEED (Leadership in Energy and Environmental Design) certification. • Lower taxes on CO₂ emissions. • Significant reduction in heating costs. • Reducing energy and water consumption. • Higher employee productivity. • Less employee sick leave. • Lower operating costs • Long-term financial benefits for companies. • In Western Europe, the cost of building passive houses is only a few percent higher, combined with a quick return on investment. • Opportunities to subsidize the construction of green buildings (subsidies, loans, support programs). • Return on investment in 5–15 years, with an average of 6 years. • Development of the construction sector. • More opportunities for sustainable construction outside the city – migration of people from the city to the countryside. • Promoting better health and well-being of the community. • Esthetics and functionality of construction. • Favorable environmental conditions in Poland, for example for the construction and operation of passive houses. • Improved quality of work/living environment and work/life comfort. • Improving the well-being and health of users. • Fulfilling the premise of “ideal well-being”. • Using a location away from dense development, and close to recreational areas. • Creating new trends in construction. • Changing people's lifestyles to be environmentally friendly. • Social responsibility for environmental protection. • Advancing social equality. • Increased environmental awareness of green building users. • Active involvement of local communities.

Figure 5. Technological, environmental and socio-economic benefits of sustainable construction. Source: [69,75–80,82,83].

Technological	Socio-economic
• Small number of companies in the market that undertake green building contracting activity. • Lack of universal design and construction methods (every user in different terrain, climate, etc.). • Green roofs pose fire hazards, higher maintenance requirements, shading of the photovoltaic panels due to excessive plant growth, the effects of the photovoltaic modules on green growth, the effect of improper panel height on air circulation inside the building. • In many countries, such as Poland, there is lack of technology and scarcity of indigenous materials, forcing the import of more expensive materials from Western Europe. • Using standard construction technologies for green construction. • Monopoly on the technology used by a few companies. • Incompetence of designers, manufacturers and component suppliers. • Limited number of contractors specializing in precast concrete systems. • Lack of early consulting on production and construction. • Design limitations due to transportation constraints. • Lack of a one-size-fits-all design and construction tool for MiC. • Limited access to components or their high price. • Difficulties in renovating, converting, or constructing buildings in the city. • Dominance of conventional construction practices/processes. • Monotonous type of construction and architecture e.g. MiC, concerns about architectural creativity. • Lack of sufficient cooperation between the representatives of science and R&D and practitioners. • Lack of knowledge and information flow among science, industry and the user. • Lack of educational programs on structural and architectural aspects. • Lack of employee competence in sustainable construction. • Barriers caused by urban planning of cities and suburbs.	• Higher costs than the traditional way of building. However, these investments pay for themselves within 5–15 years, on average 6 years. • The cost of passive houses in Poland, for example, is 10–20% higher than that of traditional houses. • High upfront costs during the design and certification processes. • Ambiguity of actual costs and benefits; significant differences in methodologies of evaluation. • Wrong economic thinking, namely that the lowest construction expenditures lead to the highest savings. • Custom designs are much more expensive than standard ones. • New products on the market continue to be much more expensive (e.g. windows and doors consistent with the passive standard). • Fluctuations in prices of materials. • High costs of upgrading existing buildings. • Inadequate or no subsidy programs for sustainable construction, e.g. in Poland. • The duration of green building projects is much longer. • Building bid prices may differ from final prices. • Unreliable sources for collecting cost-benefit data. • High cost of research and testing processes. • Impediments related to specific location requirements. • Lack of effective government incentives, subsidies and promotions. • Uncertainty as to the demand for and supply of products. • Lower demand for new construction. • Overinvestment or underestimation of costs. • Difficulty in achieving economies of scale and returns on a high initial investment. • Difficulties in obtaining funding for MiC projects. • High costs of logistics. • Concentrations of population in one place (e.g. city centers). • Rising real estate prices and other macroeconomic figures (unemployment, inflation). • Aging population and divergent economic status of the population. • A relatively new industry, considered a niche until recently. • Psychological barrier for investors and users. • Community resistance to change and innovation. • Lack of public confidence in sustainable construction. • The industry's conservative attitude toward conventional construction. • Concerns about the adaptability of construction and its flexibility to meet the needs of multiple generations. • Concerns about urban heritage or urban identity. • Spreading of myths about green buildings. • Lack of sufficient legal know-how, substantive knowledge. • Lack of dissemination of good construction practices, lack of knowledge transfer, insufficient education of the public. • Poor public acceptance due to suspicions about meeting quality expectations. • Inadequate policies and regulations in this scope. • Stereotypical thinking. • A choice between the advantages of the environment (the well-being of the city) and energy-efficient housing.

Figure 6. Technological and socio-economic barriers to sustainable construction. Source: [69,75–80,82,83].

A detailed analysis of the benefits and barriers of sustainable construction, divided into social, technological, and ecological factors, constitutes a database for identifying potential opportunities and threats, as well as the strengths and weaknesses of construction materials, and, thus, for conducting a SWOT-TOWS analysis (strengths, weaknesses, opportunities, threats—threats, weaknesses, opportunities, strengths) [84,85] for the main construction materials (glass, steel, wood, concrete). When comparing materials, technological, environmental and socioeconomic indicators were taken into account, such as the production factors of water, energy, material (including its consumption) and its recyclability, human resources, waste and its management, land use and biodiversity, environmental impact, pollution (water, air, noise), costs, material life cycle, construction and public safety, people and their well-being and working and living comfort, impact on local resources, local responsibility, economics and innovation [73].

Table 3. Estimation of the weights of the respective factors in the SWOT-TOWS analysis for the construction material—glass.

Internal Factors				External Factors			
Code	Strengths	Weight	Grade	Code	Opportunities	Weight	Grade
S1	100% recyclable raw material	0.1	5	O1	synonymous with modern design and prestige, interior design and architecture in the broadest sense through spaciousness, minimalism and elegance	0.15	5
S2	favorable energy balance—glass with a low-emission coating retains a large amount of solar energy in the room in the form of heat, thus preventing it from escaping outside	0.1	3	O2	the increasing popularity of large glazing in public buildings as well as private developments	0.05	4
S3	reduction in the level of noise from the outside due to the high acoustic insulation coefficient	0.1	4	O3	increasing demand for reducing energy consumption	0.15	5
S4	laminated glass protects against breakage—in the case of breakage, the glass will not shatter—greater strength, resistance to breakage	0.1	4	O4	production of new generation multifunctional glazing for modern construction industry	0.05	3
S5	high degree of light transmittance	0.1	5	O5	warm and bright interiors that do not need to be additionally heated or illuminated—this in turn translates into better well-being of users	0.15	4
S6	fairly long product life cycle, comparable to steel and concrete	0.05	3	O6	lower energy losses—lower costs	0.15	3
S7	heating glass can be used in skylight windows, skylights, or on glass roofs to eliminate the problem of snow and ice in winter—energy efficiency and full transparency	0.05	3	O7	zero-energy buildings, incorporating XXL-size glazing into the design they can already compete with concrete walls when it comes to energy efficiency, so the larger the area of glass, the greater the energy benefits	0.1	3

Table 3. Cont.

Internal Factors				External Factors			
S8	reducing the use of non-renewable energy sources	0.1	4	O8	new methods of construction with glass—in the case of south-facing glazing, combining low-emissivity glass with solar control glass (glazing selected this way eliminates the need to use energy-intensive air conditioners, which have a negative impact on the environment and high operating costs	0.1	3
S9	generating less solid waste by recycling glass	0.1	3	O9	heated glazing is an alternative to standard heating systems used in public facilities or private homes, which are usually impractical, unsightly and inefficient	0.1	2
S10	durability, strength, rigidity	0.1	4				
S11	resistance to environmental factors including temperature (tempered glass) or corrosion, high compression resistance, high chemical resistance	0.1	4				
Total weights		1.00		Total weights		1.00	
Sum of products of weights and ratings		3.9/5		Sum of products of weights and ratings		3.7/5	
Code	Weaknesses	Weight	Grade	Code	Threats	Weight	Grade
W1	high energy consumption during production and hardening	0.3	4	T1	high logistics costs, primarily of transportation	0.2	3
W2	glass without a low-emission coating causes heat loss through low thermal insulating power	0.3	3	T2	high prices of finished products	0.4	4
W3	high production costs	0.3	4	T3	disruption of cities' architecture, heritage or identity with glass construction	0.05	2
W4	fragility	0.1	4	T4	high prices of materials, including raw materials	0.3	4
				T5	are subject to natural hazards—earthquakes, vibrations	0.05	1

Table 3. Cont.

Internal Factors		External Factors	
Total weights	1.00	Total weights	1.00
Sum of products of weights and ratings	3.7/5	Sum of products of weights and ratings	3.55/5

Source: [8,34,46,53,73,75–77,79–81,84–92].

Table 4. Estimation of the weights of the respective factors in the SWOT-TOWS analysis for the construction material—concrete.

Internal Factors				External Factors			
Code	Strengths	Weight	Grade	Code	Opportunities	Weight	Grade
S1	high weathering resistance	0.1	5	O1	great architectural possibilities—freedom in the design of the body	0.2	5
S2	good sound insulation	0.1	4	O2	large and long experience in the market—the traditional method	0.2	5
S3	aerated concrete blocks have fairly good thermal insulation but slightly weaker frost resistance	0.05	3	O3	new market trend in the use of ferrocement	0.1	3
S4	resistant to fire and biological corrosion	0.1	5	O4	concrete is often traded in small local markets, resulting in less competition	0.1	2
S5	high strength and durability, high resistance to earthquakes	0.1	3	O5	supporting construction technology with semi-finished products, which reduces construction time	0.1	4
S6	extended service life with reduced maintenance requirements	0.1	3	O6	lower rate of price increase than, for example, wood, currently cheaper than wood	0.05	3
S7	waste disposal option	0.1	3	O7	increased return on investment	0.05	3
S8	use of cementitious materials containing, among others, iron-rich industrial waste, and agricultural waste (eggshells)—use of renewable recycled materials	0.1	4	O8	currently the most popular building material	0.1	5
S9	geopolymer concrete—geopolymer technology (use of lithium slag)	0.05	3	O9	economic attractiveness	0.1	3

Table 4. Cont.

Internal Factors				External Factors			
S10	long life cycle, comparable to that of steel and glass	0.1	3				
S11	autoclaved aerated concrete can be used for building all types of one-, two- and three-layer walls	0.1	3				
Total weights		1.00.		Total weights		1.00.	
Sum of products of weights and ratings		3.5/5		Sum of products of weights and ratings		4/5	
Code	Weaknesses	Weight	Grade	Code	Threats	Weight	Grade
W1	high energy consumption during production	0.2	5	T1	change in construction trends geared towards eco-friendly materials	0.2	5
W2	the need for technological breaks—e.g., accompanying natural processes—drying	0.1	3	T2	changing housing preferences, changing lifestyles to eco-friendly ones	0.2	3
W3	high CO ₂ emissions	0.2	5	T3	the risk that the production price of energy-intensive building materials may rise further in the coming years, due to steadily rising energy prices and more expensive CO ₂ emission allowances	0.2	4
W4	high production and construction costs (including, e.g., water consumption)	0.1	5	T4	subject to natural hazards—earthquakes, vibrations	0.1	3
W5	high material consumption	0.05	4	T5	high costs of CO ₂ emissions	0.2	4
W6	generating large amounts of waste	0.2	5	T6	high market (bargaining) power of mainly local suppliers	0.1	3
W7	low thermal insulation—high heat losses	0.1	4				
W8	depending on its properties, it has different strength and consistency	0.05	2				
Total weights		1.00.		Total weights		1.00.	
Sum of products of weights and ratings		3.6/5		Sum of products of weights and ratings		3.8/5	

Source: [8,34,46,53,73,75–77,79–81,84–92].

The rationale for choosing the SWOT-TOWS method was its versatility, practicality and simplicity in application [86]. The results of the SWOT-TOWS analysis carried out allow for comparison of the above-mentioned materials in terms of their best and most effective application in sustainable construction (Tables 3–6). The SWOT-TOWS analysis

takes into account technological, environmental, and socio-economic factors. The weighting scale used was in the 0–1 range (0—lowest importance of the factor, 1—highest importance of the factor), while the rating scale for the factor was in the 1–5 range (1—lowest rating of the factor, 5—highest rating of the factor).

Table 5. Estimation of the weights of the respective factors in the SWOT-TOWS analysis for the construction material—steel.

Internal Factors				External Factors			
Code	Strengths	Weight	Grade	Code	Opportunities	Weight	Grade
S1	high durability of galvanised steel	0.1	3	O1	the use of prefabricated products allows for precise installation with high accuracy	0.2	4
S2	homogeneity of structure and invariability of mechanical properties over time (possibility of very accurate calculations for and use of structures)	0.2	5	O2	experience in the market—a traditional method	0.2	4
S3	durability, tensile strength, compressive and bending strength, and high shear strength, allowing use of only small amounts of the material	0.2	5	O3	one of the most popular building materials on the market, next to concrete	0.2	4
S4	recyclability—demolition elements can be reused, or they are a valuable raw material for the manufacture of new steel products	0.2	5	O4	construction and installation are possible almost regardless of the season and climate conditions, which ensures rapid construction speed	0.1	5
S5	with proper protection and maintenance, not much vulnerability to natural hazards (earthquakes)	0.05	3	O5	many research and scientific centres dealing with the issues of metallurgy, including the application of steel—an extensive knowledge base on steel materials	0.2	3
S6	long life cycle comparable to glass and concrete	0.05	3	O6	in case of damage to the structure or termination of its operation, demolition of the structure is performed quickly and does not cause too many difficulties	0.1	4
S7	corresponds to the assumptions on which strength hypotheses and dimensioning methods are based	0.1	3				
S8	steel structures are easy to reinforce and adapt	0.1	4				

Table 5. Cont.

Internal Factors				External Factors			
Total weights		1.00.		Total weights		1.00.	
Sum of products of weights and ratings		4.2/5		Sum of products of weights and ratings		3.9/5	
Code	Weaknesses	Weight	Grade	Code	Threats	Weight	Grade
W1	high energy consumption during production	0.1	5	T1	change in construction trends geared towards eco-friendly materials	0.2	5
W2	reducing service life (raw steel—susceptibility to corrosion)	0.1	4	T2	changing housing preferences, changing lifestyles to eco-friendly ones	0.1	4
W3	high CO ₂ emissions	0.2	5	T3	the risk that the production price of energy-intensive building materials may rise in the coming years, due to steadily rising energy prices and more expensive CO ₂ emission allowances	0.2	4
W4	high production and construction costs (including, e.g., water consumption)	0.05	4	T4	subject to natural hazards—earthquakes, vibrations	0.1	3
W5	high material consumption	0.1	4	T5	high costs of CO ₂ emissions	0.2	4
W6	generating large amounts of waste	0.1	4	T6	high cost of materials and products	0.2	4
W7	anti-environmental impact	0.1	4				
W8	low heat and fire resistance	0.05	3				
W9	poor acoustic performance	0.05	3				
W10	some steels are characterised by sensitivity to shock loads, fatigue from dynamic loads, and fragility of components at low temperatures	0.05	3				
Total weights		1.00.		Total weights		1.00.	
Sum of products of weights and ratings		3.75/5		Sum of products of weights and ratings		4.1/5	

Source: [8,34,46,53,73,75–77,79–81,84–92].

Table 6. Estimation of the weights of the respective factors in the SWOT-TOWS analysis for the construction material—wood.

Internal Factors				External Factors			
Code	Strengths	Weight	Grade	Code	Opportunities	Weight	Grade
S1	low time-consuming production technology	0.01	5	O1	increasing demand for reducing energy consumption	0.05	5

Table 6. Cont.

Internal Factors				External Factors			
S2	flexible design	0.01	5	O2	innovative flexible designs	0.02	4
S3	lower CO ₂ emissions in comparison with other construction materials	0.1	5	O3	unlike concrete, does not need to be dried on site (dried wood from the sawmill is already used for construction)	0.01	4
S4	reducing energy consumption in the production process; capturing and absorbing carbon dioxide during tree growth	0.1	5	O4	lighter than concrete, so it is easier to expand the building (e.g., extra floor)	0.01	4
S5	wood is lighter than concrete and other construction materials, hence it is easy to process in the industrial process	0.01	4	O5	cheaper and lighter than concrete, which is a big advantage during transportation	0.02	4
S6	very good insulation material	0.05	5	O6	reduction in labor costs	0.01	3
S7	from 100% natural renewable resources	0.06	5	O7	improved living conditions, comfort, well-being, health considerations; wood naturally stabilises the microclimate in the room	0.25	4
S8	recycling possible	0.05	5	O8	wood can serve as a solid building structure for more than 100 years	0.25	5
S9	with appropriate preparations, wood becomes non-combustible	0.01	5	O9	economical design	0.02	4
S10	very long life cycle—resists aging processes permanently and for a long time	0.05	5	O10	better lifestyle, pro-environmental, more in harmony with nature	0.2	5
S11	among others regulates moisture and heat, creating a comfortable and stable climate in the room	0.02	4	O11	planting new forests provides opportunities for development of timber construction	0.05	5
S12	the most environmentally friendly material among steel, concrete, or glass	0.05	5	O12	a large field of research in the scope of innovation of the use of wooden structures in the construction industry	0.05	4

Table 6. Cont.

Internal Factors				External Factors			
S13	solid wood does not burn easily; wood is much more predictable in the event of a fire compared to other traditional building materials; during a fire, the structural load-bearing capacity of wood is maintained for a longer period of time	0.01	5	O13	the use of wood does not impede fire safety requirements	0.02	4
S14	fairly good acoustics (less echo)	0.01	4	O14	prefabricated wooden houses have extremely durable and long-lasting structures	0.05	4
S15	lower biodegradability	0.01	5	O15	wood can be used for building structures that are more resistant to earthquakes because they better absorb shocks than bricks or concrete	0.01	4
				O16	a very large number of applications in the construction industry	0.01	4
				O17	the most readily available construction material, timeless, popular and known since the earliest civilizations	0.05	3
Total weights		1.00		Total weights		1.00	
Sum of products of weights and ratings		2.71/5		Sum of products of weights and ratings		4.77/5	
Code	Weaknesses	Weight	Grade	Code	Threats	Weight	Grade
W1	less durability and less sound insulation	0.1	3	T1	higher cost of maintenance	0.1	4
W2	lower heat capacity; easy to heat up but cools down faster	0.2	4	T2	there is not enough wood in Europe to make large-scale timber construction possible, despite the fact that forests account for almost 40% of EU's land area	0.2	4
W3	restrictive technological requirements; wood must have the right moisture content (8–12 percent for interior wall cladding, ceilings and attics, 19 percent for logs with a minimum thickness of 15 cm)	0.1	3	T3	a rapid transition to wooden structures would require new investments in knowledge and production processes; moreover, it would be costly and time-consuming for the construction companies themselves	0.1	3

Table 6. Cont.

Internal Factors				External Factors			
W4	lower resistance to fire	0.1	3	T4	the problem of raw material availability; an uncontrollable increase in demand would undermine sustainable forest management	0.2	4
W5	felled trees no longer absorb CO ₂	0.1	3	T5	prices for wood react relatively quickly to changes in the market; if the inventory of suppliers, wooden building materials shrink, the price of wood will rise within one to two months	0.1	3
W6	not very resistant to climate and weather conditions (moisture, rain, snow—requires impregnation)	0.1	4	T6	underutilised potential for the development of wood-based construction	0.1	3
W7	use of additional environmentally harmful agents for maintenance and impregnation	0.2	3	T7	new investments and the cost of wood mean that wooden structures are still more expensive than concrete, cement and bricks; the cost of using wood is currently 5–10% higher, although this can vary from building to building.	0.1	4
W8	susceptibility to fungi and insects—shortening the life cycle of the material	0.1	3	T8	significant financial outlays for maintenance and care	0.1	4
Total weights		1.00		Total weights		1.00.	
Sum of products of weights and ratings		3.3/5		Sum of products of weights and ratings		3.7/5	

Source: [8,34,46,53,73,75–77,79–81,84–92].

Tables 7–10 present the calculated values of individual factors from SWOT-TOWS analyses as a product of the weight and the assessment of a given factor.

Table 7. Calculation matrix for glass.

Code	Weight	Grade	Value	Code	Weight	Grade	Value
S1	0.1	5	0.5	O1	0.15	5	0.75
S2	0.1	3	0.3	O2	0.05	4	0.2
S3	0.1	4	0.4	O3	0.15	5	0.75
S4	0.1	4	0.4	O4	0.05	3	0.15
S5	0.1	5	0.5	O5	0.15	4	0.6

Table 7. Cont.

Code	Weight	Grade	Value	Code	Weight	Grade	Value
S6	0.05	3	0.15	O6	0.15	3	0.45
S7	0.05	3	0.15	O7	0.1	3	0.3
S8	0.1	4	0.4	O8	0.1	3	0.3
S9	0.1	3	0.4	O9	0.1	2	0.2
S10	0.1	4	0.4				
S11	0.1	4	0.4				
Sum	1				1		
W1	0.3	4	0.12	T1	0.2	3	0.6
W2	0.3	3	0.9	T2	0.4	4	0.16
W3	0.3	4	0.12	T3	0.05	2	0.1
W4	0.1	4	0.4	T4	0.3	4	0.12
				T5	0.05	1	0.05
Sum	1						

Source: own research.

Table 8. Calculation matrix for concrete.

Code	Weight	Grade	Value	Code	Weight	Grade	Value
S1	0.1	5	0.5	O1	0.2	5	1
S2	0.1	4	0.3	O2	0.2	5	1
S3	0.05	3	0.15	O3	0.1	3	0.3
S4	0.1	5	0.5	O4	0.1	2	0.2
S5	0.1	3	0.3	O5	0.1	4	0.4
S6	0.1	3	0.3	O6	0.05	3	0.15
S7	0.1	3	0.3	O7	0.05	3	0.15
S8	0.1	4	0.4	O8	0.1	5	0.5
S9	0.05	3	0.15	O9	0.1	3	0.3
S10	0.1	3	0.3				
S11	0.1	3	0.3				
Sum	1				1		
W1	0.2	5	1	T1	0.2	5	1
W2	0.1	3	0.3	T2	0.2	3	0.6
W3	0.2	5	1	T3	0.2	4	0.8
W4	0.1	5	0.5	T4	0.1	3	0.3
W5	0.05	4	0.2	T5	0.2	4	0.8
W6	0.2	5	1				
W7	0.1	4	0.4				
W8	0.05	2	0.1				
Sum	1				1		

Source: own research.

Table 9. Calculation matrix for steel.

Code	Weight	Grade	Value	Code	Weight	Grade	Value
S1	0.1	3	0.3	O1	0.2	4	0.8
S2	0.2	5	1	O2	0.2	4	0.8
S3	0.2	5	1	O3	0.2	4	0.8
S4	0.2	5	1	O4	0.1	5	0.5
S5	0.05	3	0.15	O5	0.2	3	0.6
S6	0.05	3	0.15	O6	0.1	4	0.4
S7	0.1	3	0.3				
S8	0.1	4	0.4				
Sum	1			1			
W1	0.1	5	0.5	T1	0.2	5	1
W2	0.1	4	0.4	T2	0.1	4	0.4
W3	0.2	5	1	T3	0.2	4	0.8
W4	0.05	4	0.2	T4	0.1	3	0.3
W5	0.1	4	0.4	T5	0.2	4	0.8
W6	0.1	4	0.4	T6	0.2	4	0.8
W7	0.1	4	0.4				
W8	0.05	3	0.15				
W9	0.05	3	0.15				
W10	0.05	3	0.15				
Sum	1			1			

Source: own research.

Table 10. Calculation matrix for wood.

Code	Weight	Grade	Value	Code	Weight	Grade	Value
S1	0.01	5	0.05	O1	0.05	5	0.25
S2	0.01	5	0.05	O2	0.02	4	0.8
S3	0.1	5	0.5	O3	0.01	4	0.04
S4	0.1	5	0.5	O4	0.01	4	0.04
S5	0.01	4	0.04	O5	0.02	4	0.08
S6	0.05	5	0.25	O6	0.25	4	1
S7	0.06	5	0.3	O7	0.25	4	1
S8	0.05	5	0.25	O8	0.25	5	1.25
S9	0.01	5	0.05	O9	0.02	4	0.08
S10	0.05	5	0.25	O10	0.2	5	1
S11	0.02	4	0.08	O11	0.05	5	0.25
S12	0.05	5	0.25	O12	0.05	4	0.20
S13	0.01	5	0.05	O13	0.02	4	0.08

Table 10. *Cont.*

Code	Weight	Grade	Value	Code	Weight	Grade	Value
S14	0.01	4	0.04	O14	0.05	4	0.2
S15	0.01	5	0.05	O15	0.01	4	0.04
				O16	0.01	4	0.04
				O17	0.05	3	0.15
Sum	1				1		
W1	0.1	3	0.3	T1	0.1	4	0.4
W2	0.2	4	0.8	T2	0.2	4	0.8
W3	0.1	3	0.3	T3	0.1	3	0.3
W4	0.1	3	0.3	T4	0.2	4	0.8
W5	0.1	3	0.3	T5	0.1	3	0.3
W6	0.1	4	0.4	T6	0.1	3	0.3
W7	0.2	3	0.6	T7	0.1	4	0.4
W8	0.1	3	0.3	T8	0.1	4	0.4
Sum	1						

Source: own research.

As Table 7 shows, the highest value in the SWOT-TOWS analysis for glass falls on factors W2, T1, O1, O3 and O5. Therefore, glass without a low-emission coating causes heat loss through low thermal insulating power, has high logistics costs, primarily of transportation, is synonymous with modern design and prestige, interior design and architecture in the broadest sense through spaciousness, minimalism and elegance, has increasing demand for reducing energy consumption, and creates warm and bright interiors that do not need to be additionally heated or illuminated—this, in turn, translates into better well-being of users.

As can be seen from Table 8, the highest value in the SWOT-TOWS analysis for concrete falls on factors O1, O2, T1, W1, W3, W6. Therefore, respectively, concrete has great architectural possibilities—freedom in the design of the body, large and long experience in the market—is the traditional method and construction trends are changing towards eco-friendly materials, has high energy consumption during production, produces high CO₂ emissions, and generates large amounts of waste.

As shown in Table 9, the highest value in the SWOT-TOWS analysis for steel falls on the S2, S3, S4, and W3 factors, i.e., homogeneity of structure and invariability of mechanical properties over time (the possibility of very accurate calculations for and use of structures), durability, tensile strength, compressive and bending strength, and high shear strength, allowing the use of only small amounts of the material, recyclability—demolition elements can be reused, or they are a valuable raw material for the manufacture of new steel products—and high CO₂ emissions.

As can be seen from Table 10, the highest value in the SWOT-TOWS analysis for wood falls on factors O6, O7 and O10. Therefore, respectively, wood achieves a reduction in labour costs, improved living conditions, comfort, well-being, and health considerations; wood naturally stabilises the microclimate indoors, has a better lifestyle, is pro-environmental, and is more in harmony with nature.

By analysing the above tables, the following can be noted:

- The highest number of strengths was identified for wood (15) and the lowest for steel (8), while the highest number of weaknesses was noted for steel (10) and the lowest for glass (4). In contrast, the most numerous opportunities for development are for wood (18) and the least numerous for steel (6). The highest number of risks was recorded for wood (8) and the lowest for glass (5).
- The largest sum of products of strengths is for steel (4.2), and the smallest for wood (2.71—among others, this is due to the very large number of factors identified), while the highest number of weaknesses is for steel (3.75) and the lowest is for wood (3.3). As for opportunities, the largest sum of products is undoubtedly for wood (4.77) and the smallest is for glass (3.7). The risks in the largest number of sums of the products of weights and ratings are for steel (4.1) and the smallest is for glass (3.55), as shown in Table 11.
- Glass is characterised by having the greatest number of strengths and opportunities; concrete, of weaknesses and opportunities; steel, of strengths and threats; and wood, of weaknesses and opportunities.
- The most important strengths identified for glass are, first and foremost, 100% recycled raw material and a high degree of light transmittance. Weaknesses include high energy consumption during production and hardening. The most important opportunities are new methods of glass production that retain energy inside buildings and the architectural popularity of glass. The most important risks of glass include the generally high costs of production and logistics.
- The most important identified strengths of concrete are, first and foremost, high weathering resistance and corrosion resistance. The weaknesses include high energy intensity and harmful emissions. The most important opportunities are many architectural possibilities and a long tradition of use. The most important threats to concrete are the shift in construction trends to eco-friendly ones.
- The main strengths identified for steel are durability, strength and recyclability. Weaknesses include energy and emissions intensity as well as susceptibility to corrosion. The most important opportunities are all-weather performance and installation. Among the most important threats to steel is the change in construction trends to pro-environmental ones.
- The most important strengths identified for wood are mainly good thermal insulation, a long life cycle and renewability of resources. Weaknesses include low resistance to climatic conditions and lower heat capacity. The most important opportunities are structural soundness and improved living conditions. The most important risks of wood include the problem of raw material availability and higher investment costs compared to other construction materials.
- The analysis showed that the best construction material for sustainable construction is glass, due to the fact that it is a 100% recyclable raw material, and wood, due to the fact that it comes from renewable sources, stabilises the microclimate, and emits less carbon dioxide into the environment with low energy consumption. In turn, the material that does not perform well on its own in green construction is steel, due to its energy intensity and emission of harmful substances into the environment.
- Due to the specific character of each of the materials in question, their properties and manufacturing technology, as well as their established position in the construction market, the number and selection of factors for the SWOT-TOWS analysis varies. The determination of their significance and their evaluation are subjective (typical for SWOT).

Table 11. Summary of the number of factors in the SWOT-TOWS analysis for each construction material, with the sum of the products.

Material	Number/Sum of Products	Strengths	Opportunities	Weaknesses	Threats
Glass		11/3.9	9/3.7	4/3.7	5/3.55
Concrete		11/3.5	8/4	8/3.6	6/3.8
Steel		8/4.2	6/3.9	10/3.75	6/4.1
Wood		15/2.71	18/4.77	8/3.3	8/3.7

Source: own research based on [84].

The results of the SWOT-TOWS analysis classify the construction materials into one of four types, [93–95], namely:

- Aggressive (maxi–maxi) strategy, consisting of strong expansion and diversified development; it includes activities that take advantage of assets and opportunities generated by the environment.
- Conservative (maxi–mini) strategy, which involves maximizing the potential while minimizing the negative impact of the environment.
- Competitive (mini–maxi) strategy, which involves eliminating weaknesses while taking advantage of the emerging opportunities.
- Defensive (mini–mini) strategy, which involves eliminating weaknesses and avoiding threats.

As can be seen from Figure 7, glass has the most strengths and the fewest threats, concrete has the most opportunities and the fewest weaknesses, steel has the most strengths and the fewest weaknesses, and wood has the most opportunities and the fewest strengths. The most strengths are for steel, the most opportunities for wood, the most weaknesses for steel, and the most threats for steel.

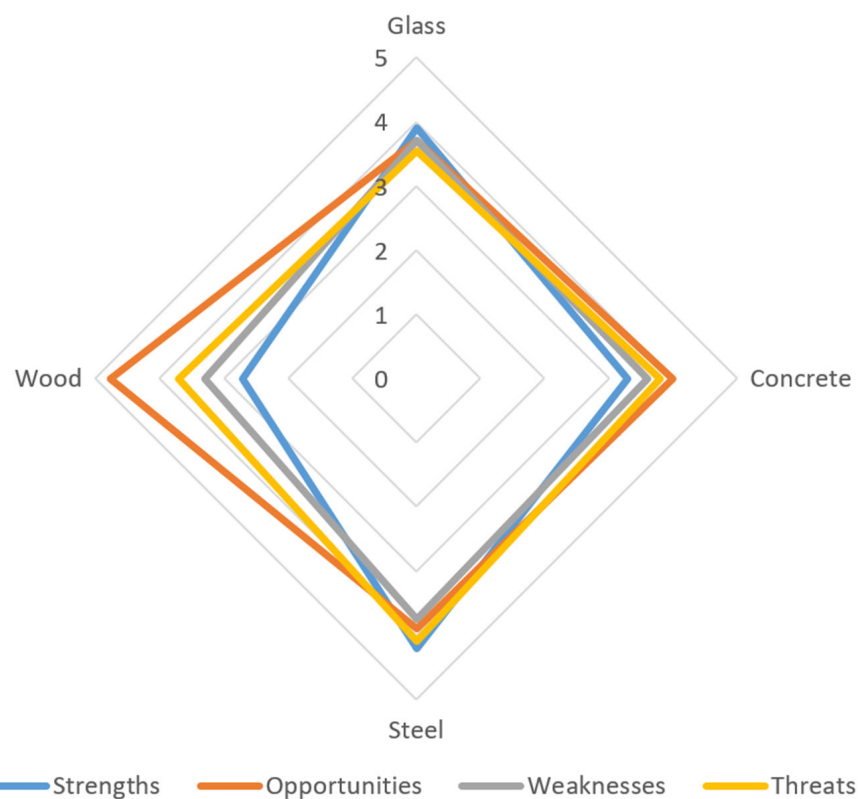


Figure 7. SWOT-TOWS analysis results for construction material.

A graphical representation of the aforementioned strategies for the structural materials analysed is shown in Figure 8.

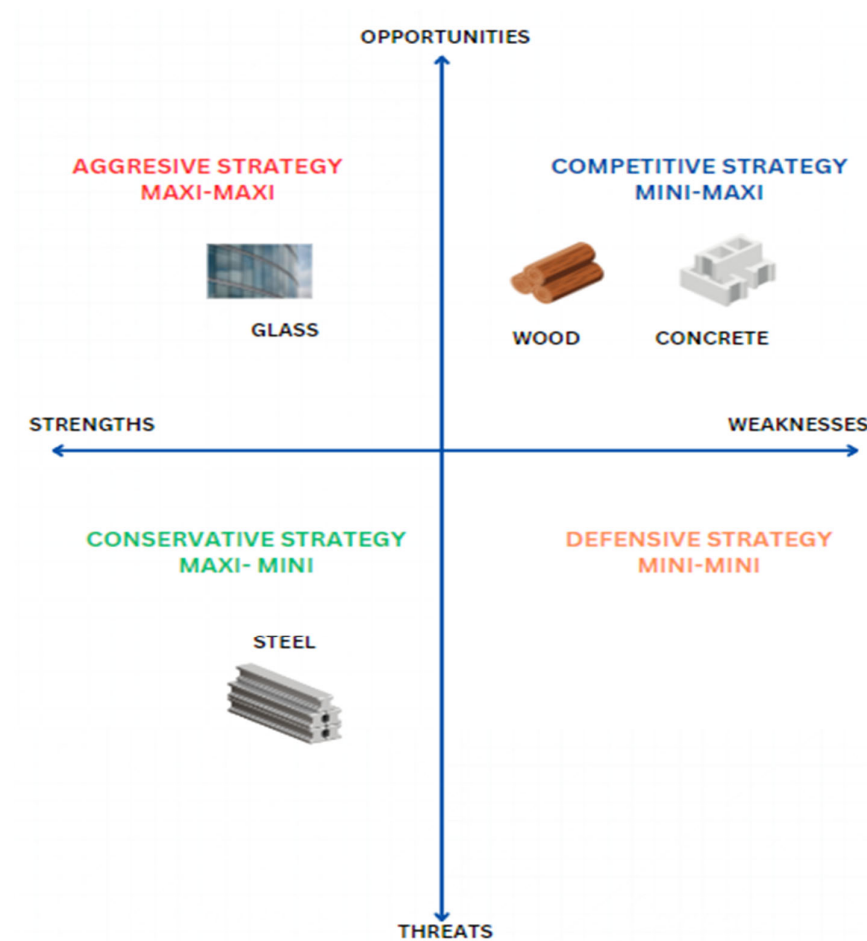


Figure 8. Graphic interpretation of the strategy in the SWOT–TOWS analysis for construction materials, such as wood, glass, concrete and steel; own research based on [84].

Based on the analysis, the following strategy has been adopted for the selected construction materials to develop sustainable construction:

- Glass—aggressive strategy—focusing on the strengths and their further development, such as glass recycling, favorable energy balance, good sound insulation, high light transmittance, resistance to environmental factors, production of heating glass, durability and strength, and opportunities, including new energy-efficient methods of glass production, and the fact that glass has now become synonymous with modern design and prestige (very high popularity of glass).
- Steel—conservative strategy—focusing on the strengths of durability, strength, and recyclability of steel, and leveling the risks, mainly reducing the costs related to CO₂ emissions and production.
- Wood—competitive strategy—focus on eliminating the weaknesses of wood, such as low fire resistance and resistance to climatic factors, and developing the opportunities, including improving living conditions, structural soundness, planting new forests, and developing research on innovative wood construction.
- Concrete—competitive strategy—focus on eliminating the weaknesses, including reducing technological breaks by changing the properties of concrete, reducing waste, reducing CO₂ emissions and reducing energy consumption for production, and devel-

opening opportunities, such as supporting construction with semi-finished products and taking advantage of the economic attractiveness of concrete.

To summarise the section of the paper on the advantages and disadvantages of sustainable construction, barriers and benefits, strengths and weaknesses, opportunities and threats for selected construction materials, it should be noted that, due to the rapidly growing green building sector, these areas are bound to change. Hence, the issues considered in the article are still an open topic and require further research. In addition, in the near future, the presented SWOT analysis may also be slightly modified, taking into account the changing conditions of green construction, especially since this method is characterised by substantial subjectivity in the selection and evaluation of criteria and factors.

A SWOT analysis conducted this way can help companies develop strategies that take into account both internal and external factors influencing the construction materials market.

6. Summary

Based on the literature review that served to prepare the SWOT-TOWS analysis and the analysis itself, observations were made regarding barriers in sustainable construction. The social barriers to the transition to energy efficient construction can be distinguished, such as the lack of knowledge—legal, know-how, content and pros and cons and dissemination of good construction practices [80]—as well as barriers caused by urban planning of cities and suburbs, an aging population and impoverishment, creating less demand for new construction (socio-economic conditions of residential construction in Poznań [in Polish]) [96]. In many countries, trends related to reducing energy intensity dominate the current construction industry. Many building solutions are based on regional traditions, habits, and weather conditions, as well as on economic aspects related to the production of building materials and their subsequent operations. Nowadays, there are many options, especially technological, convincing investors to take measures to improve the energy efficiency of buildings. The construction industry is constantly evolving, and every year, there are more and more innovative solutions that improve the energy efficiency and functionality of buildings. Modern houses are often cheap and easy to maintain, functional, easy to furnish and, moreover, economical. The practical nature of building takes into account modern solutions, such as recovery of heat from recuperation, solar panels, photovoltaics, low-emission materials, and remote control systems for all installations in the house. The increasingly popular passive or energy-efficient houses (the difference between these two types of houses is due to the materials used and their thickness) have low energy requirements for heating the interiors. Both technologies are environmentally friendly. Concrete is often used as a construction material for energy-efficient and passive houses. In the near future, it is likely that people will prefer plus-energy houses, i.e., houses that not only have negligible operating costs [75] but also produce surplus energy.

- Modern construction is often based on building with innovative materials and solutions; however, for buildings to be energy efficient, they must meet specific technical conditions. The use of modern building technologies and materials with favorable properties involves the application of those that have a low heat transfer coefficient and provide good thermal insulation. This is undoubtedly associated with higher construction costs, but in the long run, achieving the expected energy standard of the building reduces operating costs.
- However, the lack of legal, economic, and ecological knowledge, know-how, and substantive knowledge of the advantages and disadvantages of new technologies and the dissemination of good construction practices means that, unfortunately, in many regions, this type of construction is not common these days.

- In the energy-efficient construction sector, there are many opportunities to implement innovative technologies and make economically [75] and practically viable decisions to reduce the energy intensity of a building as much as possible. The energy-efficient construction sector is now forcing huge changes in the existing legislation of construction activities (implementation of Directive 2010/31/EC on the energy performance of buildings, which introduces strict energy efficiency standards from 1 January 2021) through the introduction of the Energy Efficiency Act in Poland. European Union member states, including Poland, are obliged to promote in their markets buildings that are characterised by very low energy consumption, i.e., with very high energy performance, characterised by the use of the most efficient installation and construction technologies and the use of energy from renewable sources, producing it on site or close to the building (Social and Economic Conditionings of Knowledge Transfer in Passive Building Sector [in Polish], Graczyk A.M.) [80].
- The main economic and social barriers in the construction sector to the construction of energy-efficient homes include, in particular, inadequate knowledge transfer, lack of knowledge updating, and errors in communication and cooperation between construction companies and investors. There is also a lack of cooperation and sharing of experience and of widely available information on the technologies applied, of know-how and of the dissemination of good construction practices. In general, passive construction is considered to be several tens of percent more expensive than traditional construction [75], although this is not true. In addition, for the investor, the benefits of a passive building investment result in a higher market value of the building and a reduction in heating bills over the lifetime of the building.
- The paper (Social and Economic Conditionings of Knowledge Transfer in Passive Building Sector [in Polish], Graczyk A.M.) [80] identifies barriers, divided into social (directly related to the problems of communication in the company providing construction or installation services) and economic barriers (resulting from the lack of adequate environmental education of passive construction market players)
- Social barriers include interpersonal barriers, barriers due to a lack of awareness and communication skills, barriers due to a lack of communication channels, barriers related to the quality of information, barriers related to the communication atmosphere and organizational culture of the company or institution, discrepancies of intentions, and emotions.
- Among the most important barriers related to the transfer of information are those related to its quality. Any delays and discrepancies in the information provided and received can affect the timing and execution of the project, as can divergent intentions, lack of precise performance criteria and procedures, or undefined division of responsibilities.
- In general, interpersonal barriers are related to the perception of situations through the prism of stereotypes and a lack of experience. In order to perform passive house installations, new skills are needed, and entrepreneurs do not always direct their employees to appropriate training and do not always use knowledge gained from the experience of other companies. When making the decision about a passive design, an investor is not necessarily aware that the company has no experience in building such houses. Such a lack of knowledge of the principles of passive architecture at a later stage can lead to complaints and disputes. Sometimes, there are also barriers in the construction sector due to a lack of communication channels or a negative atmosphere in the organization. The fault may lie here with both the supervisor and the employee, who, for example, fail to perform tasks beyond their standard responsibilities. Lack

of motivation, fear, and lack of confidence—these are factors that do not promote adequate employee engagement, which affects the quality of work.

- When building energy-efficient homes, economic barriers also sometimes arise. An investor cannot always afford to spend tens of percent more than for a traditional house [75]. Typically, building energy efficiently means using better insulating and quality materials; thus, to be able to compare the exact cost of building a passive building with a traditional one, one would have to focus on the same standard of construction, finishing, etc. It also happens that construction is overinvested or underestimated as a result of incomplete investor knowledge or the contractor's mistakes.
- Other important factors influencing the investment and often posing a barrier include a lack of standardised solutions (custom architectural designs are much more expensive than standard ones [75]) or the availability of components (unfortunately, the availability of many of the materials necessary for the construction of a passive building is limited).
- In conclusion, because the energy-efficient construction sector is a fairly fresh branch of the construction market, even treated as a niche until recently, that sector is now becoming more popular, especially after the adoption of the Energy Efficiency Act. Companies that, until recently, provided services to customers interested in prestigious, and, therefore, expensive, green solutions, now, often thanks to national support instruments, have the opportunity to also be available to the average investor (individual household, not just a company or municipality).

Barriers caused by urban planning of cities and suburbs, an aging population and impoverishment reduce demand for new construction (socio-economic conditions of residential construction in Poznań [in Polish]) [96].

1. Barriers caused by urban planning of cities and suburbs, an aging population and impoverishment reduce demand for new construction.
2. In the context of demographic change and the challenges of sustainable development, housing is of crucial importance, especially since in many countries, including Poland, the exodus of people from the cities to the suburbs has been very significant in recent years [96].
3. That is where the opportunities for sustainable construction come in. For many residents, owning their own home in the suburbs has become not only a symbol of social advancement and peace and quiet from the hustle and bustle of the city, but has also created an opportunity to level the playing field, especially as the problem of commuting to work or school has been solved by the spread of cars. In addition, more and more multi-family buildings are being built in the suburbs, which could be the opportunity for developers and builders to use sustainable construction.
4. However, for many people living in cities, building or renovating a dwelling in terms of the sustainable green order is an insurmountable barrier. Housing construction in city centres targets a specific audience, often families with young children, singles, the elderly and twenty- and thirty-year-olds pursuing a career or studying. The housing offered to customers should meet certain requirements. Spatial infrastructure planning must also take into account demographic changes. Noteworthy is the fact that builders are now constructing higher-standard apartments and condos that include energy-efficient construction elements.
5. Today, in many European cities, the housing stock is made up of large-panel estates that were constructed in the late 20th century. Thanks to thermal efficiency improvement and restoration of the surroundings (green areas, playgrounds), as well as due to the lack of valuable alternatives, such estates represent a specific segment of the housing market. In many areas, especially city centres, residents occupy units that are

renovated, often historic, tenement houses. At the same time, the technical conditions of old buildings are not satisfactory, often failing to meet basic standards. Only between several and a dozen old tenement houses undergo comprehensive renovation each year, a fraction of what is needed. Infrastructure deficiencies, of course, affect not only individual apartments but also sometimes entire neighborhoods.

6. The next barrier to energy-efficient redevelopment that is insurmountable in urban settings is unoccupied housing. In principle, the problems concern not only apartments where there is a frequent change of tenants (rentals) or inheritance proceedings, but also functional or service apartments and those included in the stock of the municipality. No one invests in these kind of premises, because their future status is unknown.
7. Another reason why the development of home upgrades in terms of energy efficiency may be insufficient is the socioeconomic status of the population. People with college degrees mostly have better-paying jobs and have more responsible functions within society.
8. These individuals also exhibit certain location preferences and have greater environmental awareness.
9. Such people are more likely and willing, including for financial reasons, to build passive and energy-efficient houses. Unfortunately, a huge percentage of the population has a different economic predisposition, lower-paying jobs and generally an inferior social status, which makes modern implementations in their dwellings difficult.
10. At the same time, young people often choose residential units in city centres for their own convenience. This is due to the accessibility of the city's so-called goods, such as proximity to stores, schools, swimming, sports and recreation centres, widespread and easy transportation, cinemas, etc. The choice between the advantages of the environment and living in an energy-efficient home of one's own is not so obvious.
11. In conclusion, the lack of accessibility to many utilities, financial status, and the rising prices of materials are barriers that are difficult to overcome when building passive houses (in particular for young people). In addition, the infrastructural development of cities is often one of the primary determinants of quality of life, but it does not allow for energy-saving improvements in multi-family buildings and apartment blocks, which are typical for many urban centres.
12. In many countries, the current construction industry is dominated by trends related to a variety of factors. Many building solutions are based on regional traditions, habits, and weather conditions, as well as on economic aspects related to the production of building materials and their subsequent operation.
13. The technology of housing construction has a significant impact when considering the economic aspect for the investor. The future homeowner has several alternatives to traditional technology. Already at the initial stage of planning, the investor should predict the square footage, the number of floors and the technology for constructing the building, since it is these factors that affect the cost of the investment, the execution time and the durability of the building. In addition to the still most popular masonry technology, there are also modular technology, polystyrene house construction or timber frame houses/log cabins.
14. Traditional construction in many countries (including Poland) uses masonry technology in erecting buildings. Despite the fact that other, perhaps more modern technologies, are gaining popularity, there is no indication that traditional architecture will cease to be the most popular in these regions. Traditional construction refers to brick houses, which usually have reinforced concrete foundations, masonry walls made of hollow blocks, bricks or other blocks, and a roof that is usually two- or

four-sided, constructed on the basis of a wooden roof truss (depending on how the construction work is carried out, flat roofs are also used). Of course, traditional construction also has alternatives in the form of frame or modular technology that uses prefabricated components.

15. The most commonly used building materials are autoclaved aerated concrete, ceramic, and sand lime silicates. Traditional architecture also uses traditional footings, and foundation walls are built of concrete or reinforced concrete or possibly a foundation plate.
16. Innovative technologies also include polystyrene blocks filled with concrete as a building material. Easy installation of the respective blocks, which is a building material with a high degree of thermal insulation that does not require additional insulation, such as styrofoam, is a big advantage of this method.

7. Conclusions

In light of regulations and environmental challenges, particularly climate neutrality by 2050, many industries, including the construction sector, face the challenge of incorporating sustainable practices into their projects so that the project, once implemented, is not only economically beneficial but also has a positive impact on the environment and society [97]. Numerous studies and analyses are being undertaken around the world to estimate the impact of various factors on carbon emissions, construction waste, and project costs [46]. Sustainable construction is now one of six markets with the potential for innovation, fostering competitiveness and job creation in the European economy. It aims to increase the energy efficiency of buildings, the use of building products that meet environmental criteria and clean technologies. However, it is worth remembering that it begins with an analysis of the needs of the investor and users of the building, and ends with the commissioning of the building, settlement and completion of all work, and the dissolution of organizational structures related to the execution of the project [81]. At the current stage of development of construction technology, it is not possible to completely eliminate traditional engineering materials, such as concrete, steel (reinforced concrete) or glass, due to their structural properties (strength, durability, resistance to environmental factors, such as temperature and corrosion, among others) and economic attractiveness. Therefore, for these groups of materials, the conclusions from both the SWOT analysis (the first three conclusions) and the literature review are presented below.

1. As a result of the SWOT-TOWS analysis conducted under current conditions, the construction material that best meets the requirements of sustainable construction is glass. We have identified the greatest strengths and opportunities for further development in this area. The construction materials that recorded more opportunities than threats but fewer strengths than weaknesses are concrete and wood. None of the analysed materials exhibited a predominance of weaknesses and threats. Steel ranks in a position where strengths and threats dominate. The most developmental strategy, which focuses on strengths and growth opportunities, is the aggressive strategy, which is attributed to glass. On the other hand, the conservative strategy pertains to steel, while the competitive strategy concerns wood and concrete.
2. Based on the literature and SWOT-TOWS analysis, the following future research directions can be identified:
 - the costs of sustainable construction;
 - new trends in sustainable construction in changing socio-economic conditions;
 - survey studies directed at society regarding the lack of interest and social concerns in this area.

The following limitations can be identified:

- lack of empirical research;
 - restrictions regarding the choice of available construction materials;
 - limited research on the effectiveness of using construction materials in the long-term process;
 - article length restrictions;
 - access to specific quantitative data;
 - difficulties in comparing costs and benefits in a measurable aspect;
 - possibilities for funding investments dedicated to sustainable construction, grants, etc.;
 - availability of materials and legal regulations (e.g., wood).
3. The limitations of SWOT analysis boil down to the analysis of four construction materials—wood, concrete, steel, and glass—in terms of social, technological, and environmental factors. Other aspects and materials were not analysed. Furthermore, the analysis was based on literature data from the subject literature database, Scopus.
 4. Filling the information gap, which has both scientific and practical dimensions in the field of sustainable construction, the authors attempted to address social concerns primarily arising from ignorance related to environmental, economic, and technological issues. Identifying the research gap is critical for the further development of modern solutions in the construction industry. Research in the field of sustainable construction focuses on designing and building structures that will be used in an environmentally friendly manner. Due to the multitude of factors affecting the construction industry in this matter, the authors have reviewed selected materials and technological possibilities on this topic; however, this is an issue that will be developed and continued in further publications.
 5. The study's findings contribute to the academic discourse and practical applications listed below:
 - Didactic dimension—the article serves as a knowledge base for educational purposes.
 - Practical dimension—identifying the strengths and weaknesses, opportunities, and threats of individual construction materials can serve as a basis for the broader application of these materials in construction and the development of the industry.
 - Educational goal—expanding knowledge about ecology and raising public awareness about pro-ecological issues.
 - The article can serve as a database and subject literature for other researchers and scientists, students, and practitioners.

To sum up, modern ways of building houses make it easier to construct energy-efficient and low-cost buildings. Excellent thermal insulation, as well as the use of collectors, solar panels, heat pumps, recuperation and a system to control the installation, allow us to create houses that are functional and comfortable for daily use.

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